



LEXER

Compilers | Group 05

Team 07

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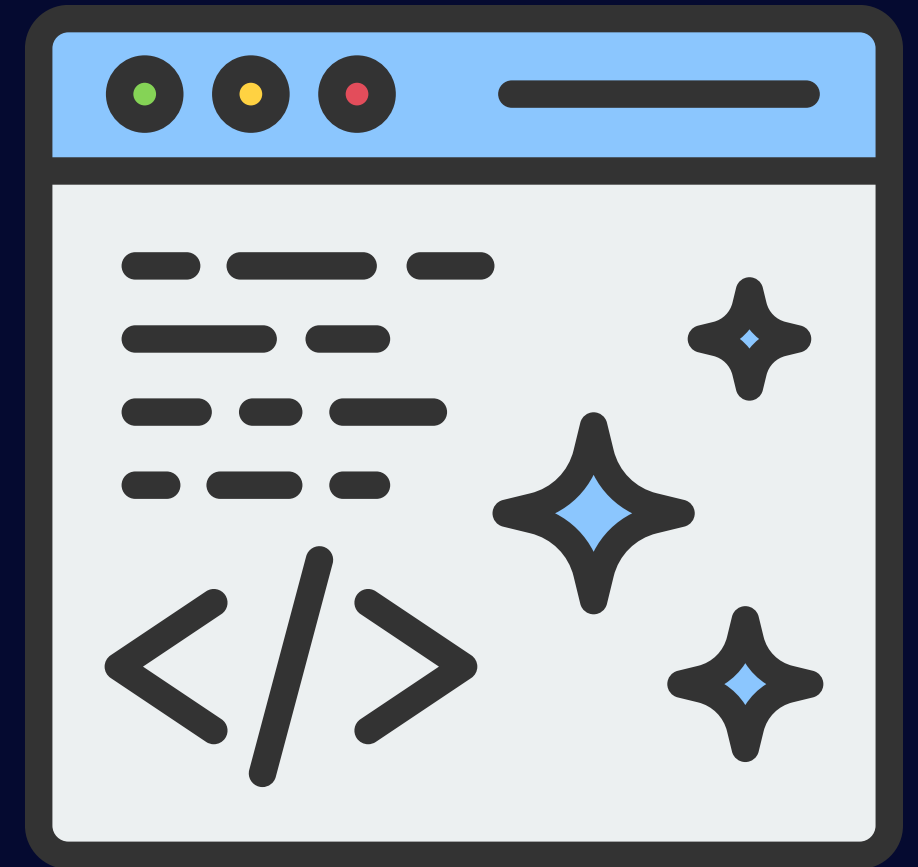
THE LEXER

06

PROOFS

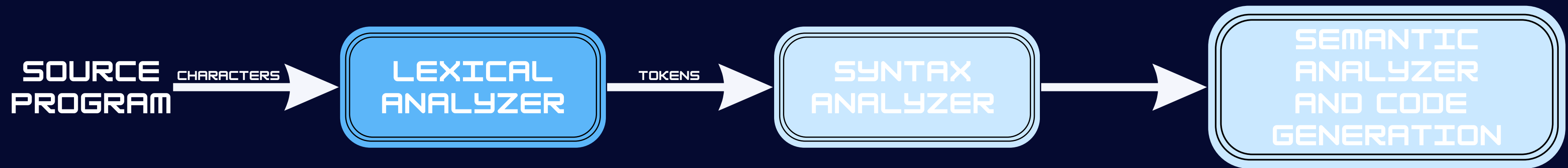
1 COMPILER

A compiler is a type of computer program that converts code from one programming language (***the source language***) into another programming language (***the target language***).



2

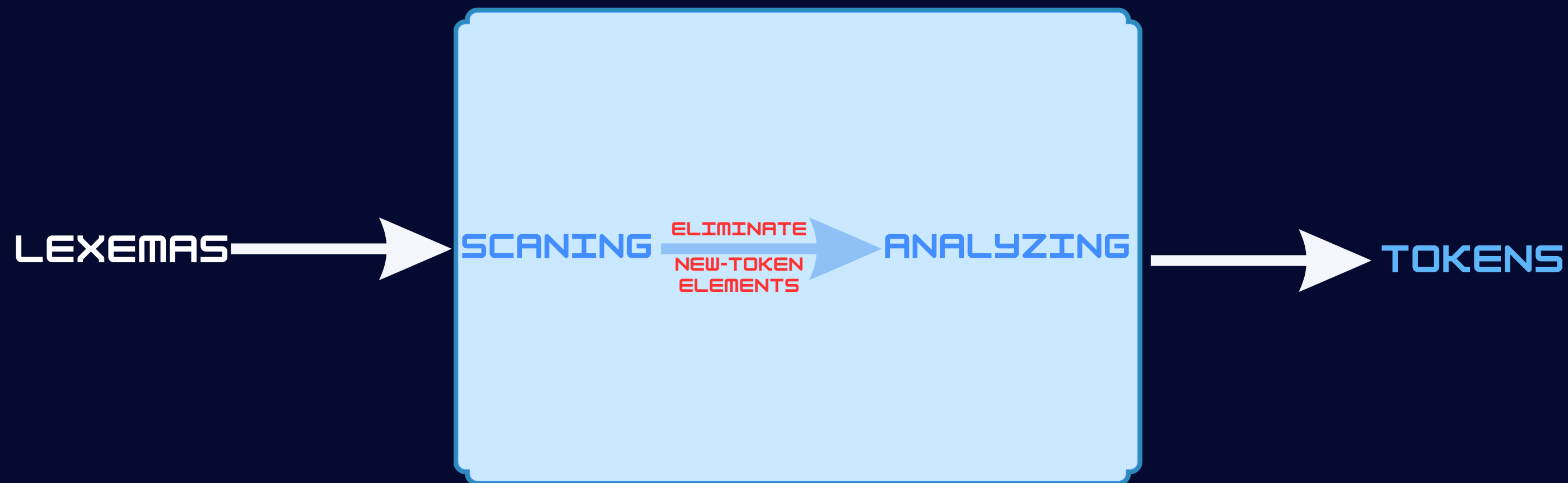
PHASES





THE LEXER

Is a computer program that takes a raw stream of characters (like source code) and breaks it down into small, meaningful pieces called tokens.



4

DETAILS

OUR TOKEN CLASSIFICATION:

- KEYWORD `print|int|return|if|else`
- IDENTIFIER `[A-Za-z_][A-Za-z0-9_]*`
- CONSTANT `[0-9]+(?:\.[0-9]+)?`
- LITERAL `"[^\r\n]*"`
- OPERATOR `== | != | <= | >= | [=+*/<>-]`
- PUNCTUATION `[(){};:,]`

TAG — TOKENS → AUTOMATA → GRAMMAR

WHAT DO WE RECOGNIZE? → HOW DO WE RECOGNIZE IT? → HOW DO WE STRUCTURE IT?



DETAILS

LEXICAL ERRORS:

- MALFORMED NUMERIC
- UNTERMINATED LITERAL
- UNRECOGNIZED CHARACTER

123abc, 3. , 1.2.3

"Hello world

@, #, \$

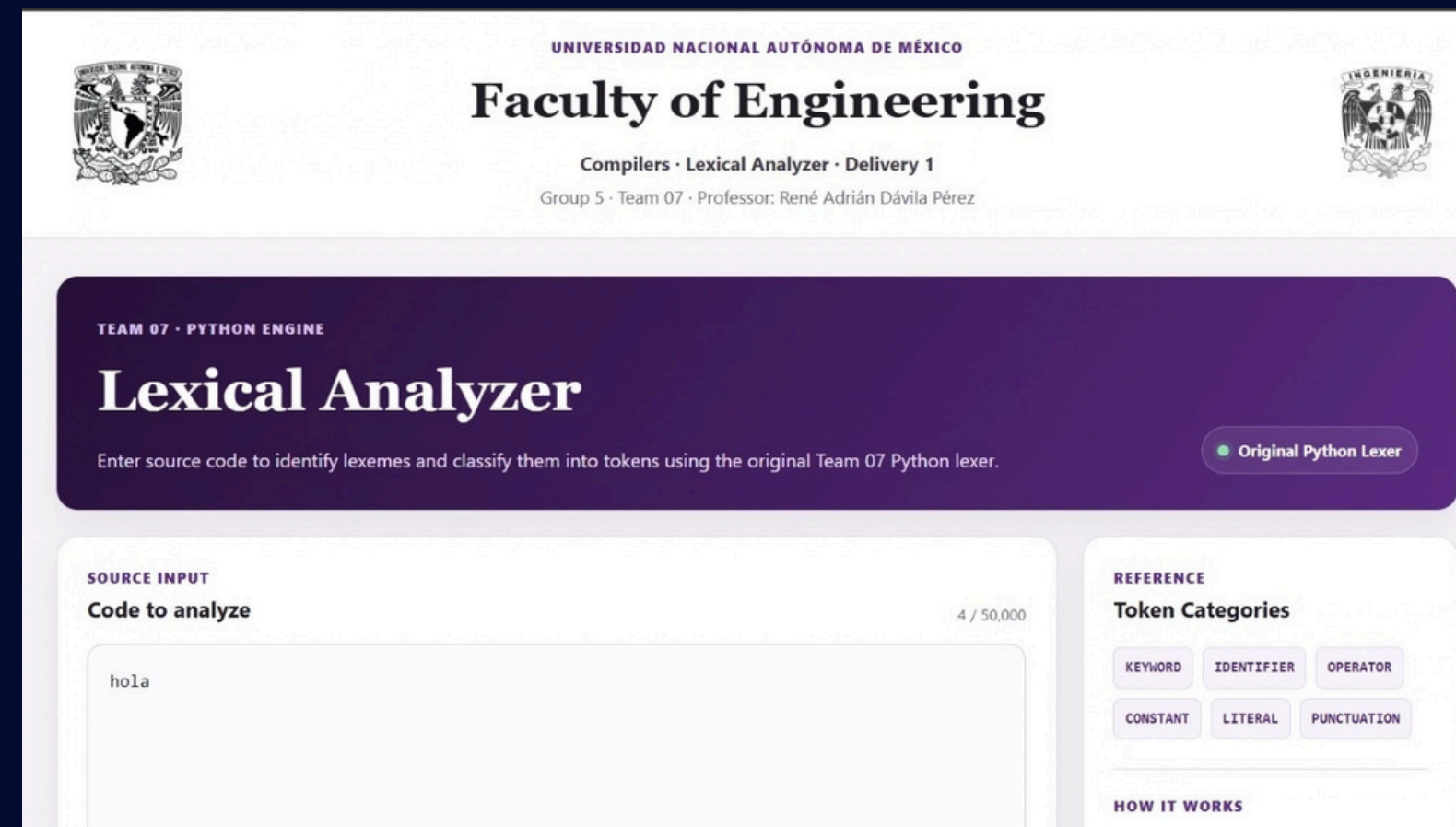
PANIC MODE

When a lexical error is detected, the Lexer reports it, consumes the problematic input, and continues scanning instead of stopping the entire analysis.

DETECT ERROR → REPORT → SKIP → CONTINUE SCANNING

5 SOLUTION

To simplify the interaction with the lexical analyzer, we implemented a web interface that provides a friendly and intuitive way to enter source code and visualize the generated tokens.





PROOFS

Magic time...

